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AM GM Inequality

Formal Definition

$$\sum_{i=1}^n \frac{x_i}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

For $x_1, x_2, \dots, x_n$.

Proof:

Consider the function $f(x) = \ln(x)$. We can see that:

$$f'(x) = \frac{1}{x} \Rightarrow f''(x) = -\frac{1}{x^2}$$

Thus this is a concave function, hence we can use the next inequality:

$$f\left(\sum_{i=1}^n \lambda_i \cdot x_i\right) \geq \sum_{i=1}^n \lambda_i \cdot f(x_i)$$

For $\lambda_i > 0 : \sum_{i=1}^n \lambda_i = 1$.

We can choose:

$$\lambda_i = \frac{1}{n} \quad \text{for } i = 1, \dots, n$$

$\Rightarrow$

$$f\left(\sum_{i=1}^n \frac{x_i}{n}\right) \geq \sum_{i=1}^n \frac{1}{n} \cdot f(x_i)$$

Replacing using the definition of $f(x)$ we have:

$$\ln \left(\sum_{i=1}^n \frac{x_i}{n} \right) \geq \sum_{i=1}^n \frac{1}{n} \cdot \ln(x_i)$$

If we apply to both sides the function $g(x) = e^x$ we preserve the inequality because the exponential function is an increasing function.

$$\exp \left[\ln \left(\sum_{i=1}^n \frac{x_i}{n} \right) \right] \geq \exp \left[\sum_{i=1}^n \frac{1}{n} \cdot \ln(x_i) \right]$$

$\Rightarrow$

$$\begin{aligned} \sum_{i=1}^n \frac{x_i}{n} &\geq \exp \left[\sum_{i=1}^n \frac{1}{n} \cdot \ln(x_i) \right] = \exp \left[\frac{1}{n} \cdot \sum_{i=1}^n \ln(x_i) \right] = \left(\exp \left[\sum_{i=1}^n \ln(x_i) \right] \right)^{\frac{1}{n}} \\ &= \left(\prod_{i=1}^n \exp[\ln(x_i)] \right)^{\frac{1}{n}} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} \end{aligned}$$

$\Rightarrow$

$$\boxed{\sum_{i=1}^n \frac{x_i}{n} \geq \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}}$$
